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Profile

Solution Architect at Nokia with a background in mathematics and statistics, currently working on AI-assisted network management, network digital twins, and real-time anomaly detection. Previous academic work includes convolution-based spatial representation analysis, explainable AI for black-box models, and coursework in self-supervised video representation learning. Interested in extending this background toward **self-supervised and predictive representation learning, video understanding, and world models**.

Professional Experience

Nokia May 2026 – Present
Solution Architect, AIOps & Network AI *Taipei, Taiwan*

- **Agentic AIOps for network management:** Serve as the technical bridge between customers and R&D for AI-assisted network management systems. Define operational use cases, translate customer troubleshooting workflows into product requirements, and validate agentic AI behaviors through end-to-end test scenarios.
- **Network digital twin:** Develop containerlab-based virtual telecom network environments for validating network automation, troubleshooting workflows, and AI-assisted network operations without relying on physical lab devices.
- **Real-time anomaly detection:** Work on anomaly detection and router health monitoring using real-time network telemetry, with the goal of identifying abnormal network behavior and providing interpretable health information to network operators.

Nokia Nov 2024 – Apr 2026
Network Engineer, AI/ML Systems Projects *Taipei, Taiwan*

- **RAG-based product knowledge system:** Built a retrieval-augmented generation system over 15 long-form telecom product knowledge sources, each approximately 50–100 pages, reducing typical information lookup time from 30–60 minutes to 10–30 seconds and achieving over 90% pass rate on 50–100 validation questions.
- **Capacity intelligence prototype:** Independently developed an AI prototype over 5.93M hourly observations from 16,482 4G cells and 223 KPIs, covering capacity forecasting, congestion-risk identification, anomaly analysis, automated root-cause investigation, and AI-assisted KPI exploration.

Ericsson Jan 2024 – Jun 2024
Master Thesis Student, Trustworthy / Explainable AI *Stockholm, Sweden*

- Conducted thesis research at Ericsson R&D on generating interpretable rule-based explanations for black-box machine learning models, using telecom HTTP-delay prediction as the application.
- Developed a post-hoc explanation pipeline that generated counterfactual samples around individual predictions, estimated their local probability distributions, and extracted human-readable decision rules using fuzzy labeling and decision trees.
- Evaluated the extracted explanations against existing rule-based explanation methods in terms of how faithfully they reproduced the original model's predictions, how often valid explanations could be generated, and the complexity of the resulting rules.

Education

National Yang Ming Chiao Tung University 2021 – 2024
M.S. in Statistics GPA: 3.96 / 4.30 *Taiwan*

- Thesis: *Spatial Sparse Principal Component Analysis Using Autoencoders with Convolutions* [Thesis] [Code].
- Studied the effect of introducing convolutional layers into sparse autoencoder-based PCA for spatially structured data, focusing on whether convolution helps preserve local spatial structure in learned sparse representations.
- Conducted 1D linear, 2D linear, and nonlinear simulation experiments and compared convolutional and non-convolutional variants with PCA, Sparse PCA, and sparse autoencoder baselines.
- Applied the framework to monthly sea surface temperature data over the equatorial Pacific Ocean from 2003–2020.

KTH Royal Institute of Technology 2023 – 2024
Exchange Program in Computer Science *Stockholm, Sweden*

- Degree project conducted at Ericsson R&D: *Explanation Analysis Using Rule Extraction* [Thesis].
- Coursework included deep learning, reinforcement learning, and image analysis and computer vision, with practical work in self-supervised learning, semi-supervised learning, and model interpretability.

National Tsing Hua University 2017 – 2021
B.S. in Mathematics *Taiwan*

Selected AI / ML Projects

Siamese Masked Autoencoder (SiamMAE) Reproduction – KTH Course Project [Code] PyTorch
- Reproduced SiamMAE for self-supervised video representation learning, using UCF-101 for pretraining and DAVIS-2017 for video object segmentation evaluation.
- Implemented masked frame prediction and label propagation to study temporal visual correspondence without dense supervision.

S&P 500 Quantitative ML Dashboard – Independent Project [Code] [Live Demo] Python, FLAML, React
- Built an end-to-end stock-selection pipeline using 40+ point-in-time features and leakage-aware expanding walk-forward validation.
- Achieved 57.3% cumulative return versus 15.3% for SPY over the 11-month final out-of-sample period; integrated backtesting and Fama–French factor analysis into an interactive dashboard.

Taiwanese Food Classification App – Course Project [Code] [Demo] TensorFlow, EfficientNetV2
- Built an EfficientNetV2S classifier for 101 Taiwanese food categories with Top-5 prediction and user correction feedback.
- Achieved 69% Top-5 accuracy on the Kaggle test set and investigated performance degradation on web-scraped images.

Deep Learning Coursework – KTH [Code] PyTorch
- Implemented SimCLR and FixMatch for self- and semi-supervised learning, together with 4+ visual attribution methods including Grad-CAM and integrated gradients.
- Studied model bias, memorization, layer criticality, and quantitative evaluation of deep-network explanations.

Camera-Based Self-Driving Robot [Code] PyTorch, OpenCV, ROS
- Built an end-to-end real-time lane-following pipeline from camera input and visual prediction to ROS-based robot control.

Awards

2nd Place, Nokia MS Espresso AI Competition 2026
Independently developed the Capacity Intelligence prototype; ranked 2nd among 100+ participants.

Gold Award, Intelligent Manufacturing Big Data Analytics Contest – 1st Place Nationwide
Led a two-person team to 1st place among approximately 200 teams with an EfficientNet-based defect classification system.

Honorable Mention, SinoPac AI GO Competition – Top 10 / 800+ Teams
Co-developed a hybrid machine-learning and rule-based rare-event prediction system; achieved a private leaderboard F1 score of 0.8825.

Skills

Programming: Python, C++, R, Git, Linux
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, FLAML, Computer Vision, Self-Supervised Learning, Explainable AI, Forecasting, Anomaly Detection
Statistics: Statistical Modeling, Simulation, Regression, Time Series Analysis, Optimization, PCA, Sparse PCA, Cross-Validation
AI Systems: RAG, Agentic AI, MCP, LangChain, FastAPI, Docker
Network Systems: Containerlab, Network Automation, AIOps, Network Telemetry, Telecom Network Management